The Hidden Materiality of Growth

Pierre Campergue's Economic Formalism and Evidence from GDP-Energy Coupling

Author: Mélik Lemariey
Status: working paper for repository publication
Date: 8 May 2026
Repository context: articles derived from Pierre Campergue's economic corpus
Empirical files: `merged_energy_gdp.csv`, `regression_summary.csv`, `rolling_loglog_elasticity.csv`, `run_metadata.json`

Abstract

This working paper develops an empirical and conceptual article from Pierre Campergue's economic corpus, especially the manuscript *Reflections on the Economic Power of Work and the Mechanical Power of Money*. It argues that Campergue's contribution should not be reduced to the statement that "economics is thermodynamics." A more accurate reading is that Campergue uses the discipline of thermomechanical formalism to clarify economic language: Cash is distinguished from Money, labour from Economic Work, and accounting circulation from productive transformation.

The empirical part tests a narrower proposition compatible with this formalism: if economic growth is not merely an accounting expansion, then real output should remain materially conditioned. Energy use is used here as a partial empirical trace of that materiality. Using World Bank World Development Indicators, the paper reconstructs an energy-use proxy with the dimensionally coherent formula `energy = GDP_PPP_constant / GDP_per_unit_energy_PPP_constant`. The current run covers the world aggregate and five countries - China, India, the United States, Germany, and France - over an effective 1990-2023 sample.

The results show strong world-level GDP-energy coupling: the world log-log elasticity is 1.605 with R^2 = 0.991, while the annual log-change model remains materially associated with beta = 0.809 and R^2 = 0.661. China and India show strong long-run coupling. France, Germany, and the United States show more complex territorial patterns. This supports a distinction between global material coupling and national territorial decoupling. The evidence does not validate Campergue's complete theoretical corpus. It supports a limited claim: global growth remains materially and energetically conditioned, while some national accounts can make this materiality less visible.

Keywords: Pierre Campergue, Economic Work, Cash, Money, GDP, energy, decoupling, productive transformation, World Bank, materiality of growth

1. Introduction

Modern economies are often described through monetary aggregates: GDP, income, value added, margins, prices, credit, market capitalization, and public accounts. This language is necessary, but it can also conceal the physical and organizational transformations that make production possible. Goods must still be extracted, transformed, transported, cleaned, repaired, maintained, powered, and delivered. Even services depend on infrastructures, equipment, logistics, data centers, buildings, devices, and global supply chains.

Pierre Campergue's corpus is valuable because it attacks the problem at the level of economic vocabulary. The manuscript *Reflections on the Economic Power of Work and the Mechanical Power of Money* does not merely claim that economics resembles thermodynamics. Its more precise contribution is methodological: economic analysis should distinguish supports from values, tasks from productive transformations, and accounting flows from real changes in productive conditions.

This paper uses Campergue's formal distinctions to frame an empirical question: does economic growth remain materially coupled to energy use? The question is narrower than Campergue's full system. It does not test the entire Work-to-Money formalism. It tests one macroeconomic implication: if growth is a real transformation rather than only a monetary inscription, then at the world scale real output should remain connected to physical energy use.

The paper combines three elements. First, it presents Campergue's conceptual distinctions: Cash versus Money, labour versus Economic Work, and transactions versus transformations. Second, it uses public World Bank data to estimate GDP-energy coupling. Third, it interprets the country-level heterogeneity as a visibility problem: some countries record monetary value where the productive energy has been consumed elsewhere.

The core thesis is:

> Campergue's distinctions between Cash and Money, labour and Economic Work, and transactions and transformations provide a rigorous conceptual language for re-reading macroeconomic growth. When that language is brought into contact with World Bank data, the current evidence supports strong world-level GDP-energy coupling and heterogeneous country-level results consistent with territorial decoupling rather than full dematerialization.

2. Campergue's formal distinctions

2.1 Cash and Money

A central distinction in Campergue's corpus is the separation between **Cash** and **Money**.

Cash is the monetary support: a sign, unit, inscription, bank balance, or circulating medium. Money is the economic value or purchasing power carried by that support. A banknote, coin, or deposit entry may remain formally identical while the purchasing power it carries changes. This means that the movement or creation of monetary units should not be automatically confused with the creation of real economic value.

This distinction is not identical to the terminology used by central banks, but it speaks to a related problem. The Bank of England explains that modern money takes different forms, including currency, bank deposits, and central bank reserves, and that most money in modern economies is created as bank deposits through commercial-bank lending. The European Central Bank also distinguishes forms of money such as cash, deposits, and central bank reserves. These official descriptions confirm that "money" is not a single simple object; it is an institutional structure made of different supports and claims.

Campergue's vocabulary goes further by asking whether the support carries new purchasing power or merely redistributes claims. A credit entry may alter Cash balances without necessarily creating new real productive capacity. This does not mean that banks or credit are useless. It means that monetary inscription and real creation should not be treated as the same operation.

2.2 Labour and Economic Work

A second distinction is between **labour** and **Economic Work**.

Labour is the task to be done: an activity, job, effort, or action performed by humans or organizations. Economic Work, in Campergue's sense, is tied to productive transformation: the modification of production conditions, costs, speed, utility, or capacity. The distinction matters because an activity can be socially useful without directly altering productive conditions, while a machine, process, or energy system can transform production without being "labour" in the ordinary human sense.

This is where the formal discipline of the corpus becomes important. Campergue does not simply attach moral labels to occupations. The analytical issue is whether an operation transforms the system or merely moves, classifies, records, allocates, or settles claims. Productive and non-productive are therefore technical categories, not moral categories.

2.3 Transactions and transformations

Conventional accounts often describe economies as flows: goods flow, services flow, money flows, income flows. Campergue's corpus asks what transformations generate those flows. A transaction can transfer ownership or settle a claim without modifying the product. A transformation changes the product, process, utility, cost, speed, or productive capacity.

This distinction is crucial for interpreting GDP. GDP records value added, but it does not always make clear where the underlying energy-intensive transformations occurred. In a global value chain, design, brand, finance, distribution, and margins can be recorded in one country while extraction, manufacturing, logistics, and energy use occur elsewhere.

The question is not whether services are "fake" or useless. The question is whether national accounts can separate the place where Cash is recorded from the place where Economic Work occurs.

---

3. Literature position

3.1 Energy and economic growth

The energy-growth literature is large and methodologically diverse. Stern's multivariate approach to U.S. energy and GDP is one early reference point. Later surveys by Payne and Ozturk emphasize that conclusions vary with country, period, variables, and econometric specification. This matters for the present paper: no single regression establishes a universal causal law.

Other work has identified stable broad relationships between income and energy use. Csereklyei, Rubio-Varas, and Stern describe stylized facts linking energy use and economic growth over long periods. Decoupling literature adds nuance: Haberl and co-authors' systematic review finds that relative decoupling is observed in some domains, but strong absolute decoupling is much harder to establish, especially when useful energy and exergy are considered.

Recent public research commentary reaches a related conclusion at the global level: apparent decoupling in rich economies can be weakened once offshoring and embodied energy are considered. The UK Energy Research Centre also reports that energy efficiency can itself contribute to GDP growth, complicating the simplistic idea that efficiency necessarily breaks the energy-growth relationship.

The implication is not that energy mechanically "causes" GDP. Rather, GDP and energy are co-determined inside the same production system. The present paper therefore estimates association and comovement, not one-way causality.

3.2 Money, production, and monetary supports

The monetary literature also helps position Campergue. Official central-bank sources describe modern money as a hierarchy of cash, reserves, and bank deposits. The Bank of England's 2014 Quarterly Bulletin notes that commercial bank deposits constitute most money in modern economies and are created through lending. The ECB similarly describes money through its functions and institutional forms.

More theoretical traditions also reject the idea that money is a neutral veil. Graziani's monetary theory of production treats money as constitutive of production, income, and distribution, while Ingham's credit theory treats money as a social relation. These approaches differ from Campergue, but they support the broader point that money must be analyzed through its institutional support, its value, and its role in production.

Campergue's Cash/Money distinction can therefore be read as a formal attempt to separate the support of monetary expression from the real purchasing power or economic value that support carries.

---

4. Data and dimensional reconstruction

4.1 Sources

The empirical analysis uses World Bank World Development Indicators obtained through the World Bank API. Three indicators are used:

Variable	World Bank code	Meaning
`gdp_real`	`NY.GDP.MKTP.KD`	GDP in constant market U.S. dollars
`gdp_ppp_real`	`NY.GDP.MKTP.PP.KD`	GDP in constant PPP international dollars
`gdp_per_energy`	`EG.GDP.PUSE.KO.PP.KD`	GDP per unit of energy use, PPP constant international dollars per kg of oil equivalent

The World Bank metadata define `EG.GDP.PUSE.KO.PP.KD` as GDP per unit of energy use expressed in PPP dollars per kilogram of oil equivalent. The underlying energy data are sourced from the International Energy Agency. The World Bank API supports CSV downloads using the `downloadformat=csv` parameter.

The current run metadata report:

```
countries: WLD, USA, FRA, DEU, CHN, IND
requested period: 1965-2023
```

```
effective merged period: 1990-2023
energy_method: reconstructed_dimensionally
energy_formula: energy = gdp_ppp_real / gdp_per_energy
observations: 204
```

The effective period begins in 1990 because the merged selected series do not provide a complete 1965-2023 overlap for the chosen indicators and countries.

4.2 Reconstruction formula

Energy use is reconstructed as:

```
energy = gdp_ppp_real / gdp_per_energy
```

This is dimensionally coherent. If:

```
gdp_ppp_real = PPP dollars
gdp_per_energy = PPP dollars per kg oil equivalent
```

then:

```
energy = PPP dollars / (PPP dollars per kg oil equivalent)
       = kg oil equivalent
```

This corrects an earlier weaker construction that divided market-price GDP by a PPP-based GDP-per-energy ratio. The corrected construction is not equivalent to using a fully independent physical-energy series, but it is internally coherent.

4.3 Models

For each country or aggregate, three OLS models are estimated.

Level model:

```
GDP_i,t = alpha_i + beta_i Energy_i,t + epsilon_i,t
```

Log-log model:

```
log(GDP_i,t) = alpha_i + beta_i log(Energy_i,t) + epsilon_i,t
```

Annual log-change model:

```
Delta log(GDP_i,t) = alpha_i + beta_i Delta log(Energy_i,t) + epsilon_i,t
```

The level model is intuitive but sensitive to common trends. The log-log model provides an elasticity. The annual log-change model is more demanding because it tests whether yearly changes move together, not merely whether two variables trend upward over time.

---

5. Results

The main results from the current run are summarized in the compact table below. Area codes are WLD (World), CHN (China), IND (India), USA (United States), DEU (Germany), and FRA (France). The full level-regression output is available in `regression_summary.csv`.

Area	N log	beta log-log	R^2 log-log	N dlog	beta dlog	R^2 dlog
WLD	34	1.605	0.991	33	0.809	0.661
CHN	34	1.659	0.979	33	0.277	0.152
IND	34	1.459	0.996	33	0.831	0.479
USA	34	2.527	0.280	33	0.567	0.655
DEU	34	-1.270	0.611	33	0.274	0.193
FRA	34	0.259	0.012	33	0.295	0.260

5.1 World aggregate

The world aggregate is the central test because the world cannot offshore production outside itself. At the world level, the relationship between real GDP and reconstructed energy use is strong in all relevant specifications.

The log-log elasticity is 1.605 with R^2 = 0.991. This means that over the observed 1990-2023 period, a 1% increase in reconstructed world energy use is associated with approximately a 1.6% increase in real world GDP. This is not a causal estimate, but it is a strong long-run association.

The annual log-change model remains meaningful: beta = 0.809 with R^2 = 0.661. This reduces the risk that the result is only a visual artifact of two trending series.

5.2 China and India

China and India show strong long-run GDP-energy coupling. China has a log-log elasticity of 1.659 with R^2 = 0.979. India has a log-log elasticity of 1.459 with R^2 = 0.996. These results are consistent with economies where industrialization, infrastructure growth, manufacturing expansion, and urbanization keep energy use and recorded output closely aligned.

India also shows a relatively strong annual log-change relation, with beta = 0.831 and R^2 = 0.479. China's annual log-change relation is weaker, with beta = 0.277 and R^2 = 0.152, suggesting that short-run variation is affected by structural change, policy, efficiency gains, and timing.

5.3 United States, Germany, and France

The United States, Germany, and France show more complex patterns.

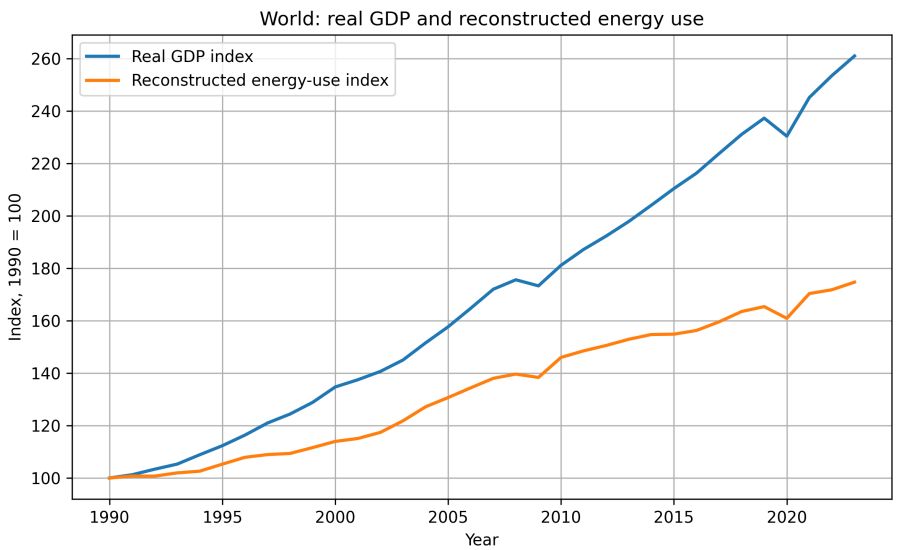

Figure 1: World indexed series

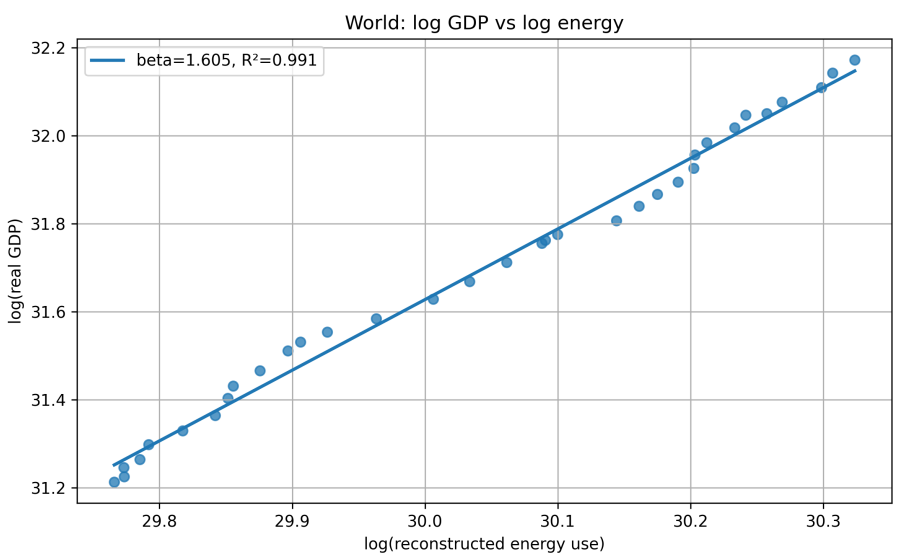

Figure 2: World log-log relation

The United States has a weak level and log-log relationship, but a relatively strong annual log-change relation: beta = 0.567 with R^2 = 0.655. This suggests that short-run GDP and energy movements remain linked, even though the long-run territorial level relationship is distorted by structural change.

Germany has a negative log-log elasticity (-1.270) with R^2 = 0.611. France has almost no log-log relation (beta = 0.259, R^2 = 0.012). These results should not be interpreted as proof that energy no longer matters. A more plausible interpretation is that territorial national statistics no longer align the location of productive energy use with the location of value added and monetary recording.

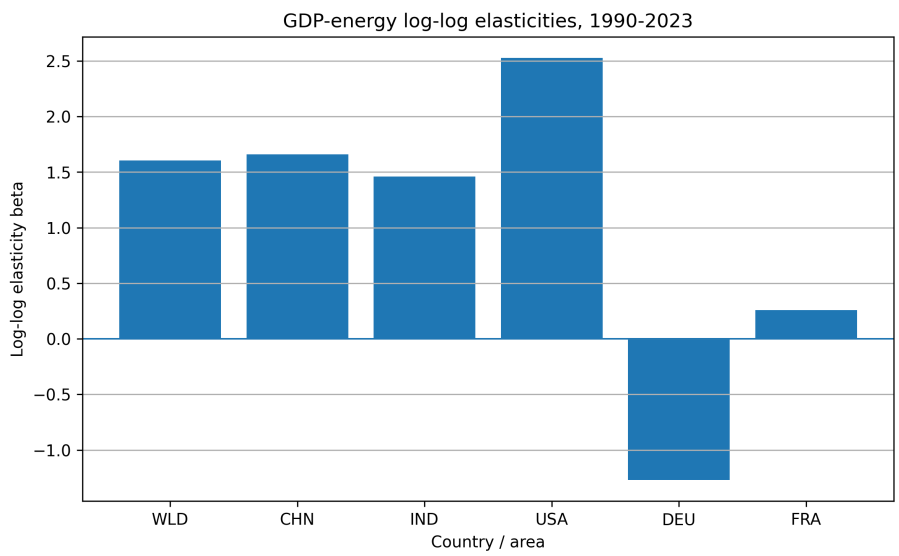

Figure 3: Country log-log elasticities

5.4 Rolling elasticity

The rolling elasticity output provides a dynamic view of the same problem. The rolling windows show that the GDP-energy relation is not fixed once and for all. It varies by country and period, as expected from the literature on energy-growth causality and decoupling.

---

6. Productive visibility, endogeneity, and the Cash/Money distinction

The country-level results should not be read as a moral ranking between productive and non-productive economies. The stronger interpretation is that they measure the degree of **territorial visibility** of productive transformation.

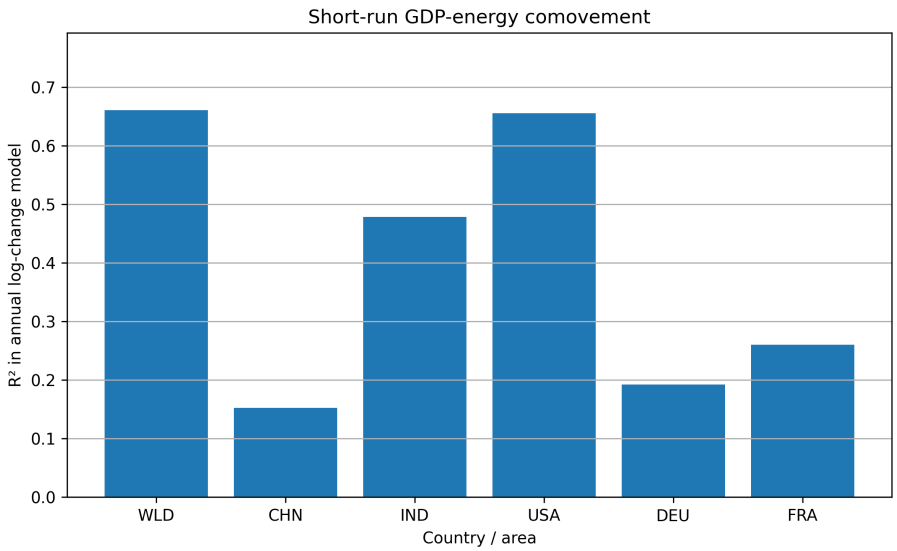

Figure 4: Annual log-change comovement

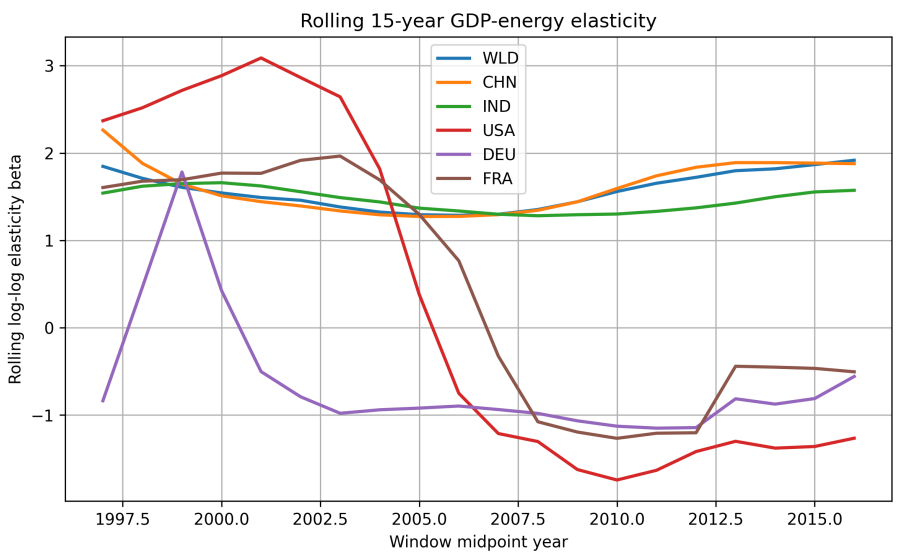

Figure 5: Rolling elasticity

In economies where energy consumption, productive transformation, and recorded GDP remain located within the same national boundary, the GDP-energy relationship remains visible. This appears in the current results for China and India.

In mature service economies, the relation may weaken or become negative. This does not necessarily mean that energy has become irrelevant. It may mean that the place where energy is consumed, the place where productive transformation occurs, and the place where monetary value is recorded have become territorially separated.

This connects directly with Campergue's Cash/Money distinction. GDP is expressed in monetary units and accounting conventions. It is a Cash-like representation in national accounts. Money, in Campergue's vocabulary, refers to purchasing power or economic value. Productive transformation, however, may occur elsewhere. A country can record monetary value domestically while the energy-intensive Work supporting that value is performed abroad.

The global result is therefore essential. At the world level, production cannot be shifted outside the accounting boundary. This is why the world-level coupling is the strongest evidence in the current note.

A useful conceptual chain is:

```
Energy inputs and technical systems
        -> Economic Work / productive transformation
        -> changes in cost, utility, speed, output, and capacity
        -> real production and value added
        -> Money / purchasing power
        -> Cash / monetary recording in accounts
```

In global value chains, this chain can be territorially split:

```
Energy and transformation in country A
        -> product, component, logistics, infrastructure
        -> design, brand, margin, finance, distribution in country B
        -> GDP and monetary value recorded in country B
```

This is why a weak national energy-GDP correlation cannot be interpreted as complete dematerialization. It may reflect a broken territorial alignment between Cash, Money, Economic Work, and energy.

---

7. Discussion

The empirical findings support three claims.

First, there is strong evidence of world-level GDP-energy coupling in the current World Bank run. The log-log association is very high, and the annual log-change model remains materially meaningful.

Second, national relationships are heterogeneous. This is consistent with the energy-growth literature, which repeatedly finds that results depend on country, period, model, and variable choice.

Third, the heterogeneity itself is informative. Countries with visible industrial and infrastructure expansion show strong coupling. Mature service economies show weaker or negative territorial relationships. This supports the idea that the materiality of growth can be hidden by national accounting boundaries, not necessarily eliminated.

The article therefore avoids two extremes. It does not claim that energy mechanically determines GDP. It also does not accept the simplistic claim that advanced economies have become immaterial. The evidence supports a middle position:

> Growth remains materially conditioned, but its material basis is increasingly displaced, mediated, and statistically obscured.

This is precisely where Campergue's formal distinctions are useful. The distinction between Cash and Money prevents the analyst from treating monetary inscriptions as real creation by default. The distinction between labour and Economic Work prevents the analyst from treating all activity as equivalent transformation. The distinction between transaction and transformation prevents the analyst from confusing accounting circulation with productive change.

---

8. Limitations

This paper is deliberately limited.

First, the energy variable is reconstructed from World Bank indicators. The reconstruction is dimensionally coherent, but a stronger robustness check should use independent physical energy series such as IEA total energy supply, final energy consumption, or comparable exajoule-based datasets.

Second, GDP and energy are endogenous. Energy enables production, but GDP growth also increases the capacity to consume energy. The regressions estimate association, not one-way causality.

Third, level and log-log regressions are sensitive to common trends. The annual log-change model reduces this problem but does not eliminate all econometric issues.

Fourth, national statistics are territorial. They do not fully capture embodied energy in imports or global supply-chain displacement. This is particularly important for advanced service economies.

Fifth, the current sample includes only six country or area codes. A broader panel would be needed to generalize the country-level interpretation.

Sixth, the empirical note does not validate Campergue's full corpus. It tests one limited macroeconomic implication compatible with the corpus.

---

9. Reproducibility

The data can be downloaded with:

```
python scripts/download_wb_sources.py --out data/code --raw data/raw
```

Then run:

```
python src/energy_gdp_analysis.py \
  --gdp-file data/code/API_NY.GDP.MKTP.KD.csv \
  --gdp-ppp-file data/code/API_NY.GDP.MKTP.PP.KD.csv \
  --gdp-per-energy-file data/code/API_EG.GDP.PUSE.KO.PP.KD.csv \
  --countries WLD,USA,FRA,DEU,CHN,IND \
  --start-year 1965 \
  --end-year 2023 \
  --out output_macro
```

The expected outputs are:

```
merged_energy_gdp.csv
regression_summary.csv
rolling_loglog_elasticity.csv
run_metadata.json
*_energy_gdp_level.png
*_energy_gdp_loglog.png
*_energy_gdp_dlog.png
*_indexed_series.png
rolling_loglog_elasticity.png
```

---

10. Conclusion

Pierre Campergue's corpus invites economists to separate accounting circulation from productive transformation, monetary support from purchasing power, and task execution from Economic Work. These distinctions are not merely semantic. They alter the way empirical growth should be interpreted.

The current World Bank run supports one limited but important proposition: at the world scale, real GDP and reconstructed energy use remain strongly coupled over the effective 1990-2023 sample. At the national scale, the relation is heterogeneous. Some economies show strong coupling; others show weak or negative territorial relationships. This is consistent with the hypothesis that advanced economies do not necessarily escape materiality; they may displace it through efficiency, trade, offshoring, services, and global value chains.

The evidence does not prove Campergue's complete theoretical formalism. It does support the need for the kind of conceptual discipline his corpus demands. Economic growth should not be read only through monetary signs. It should also be read through the transformations that make those signs meaningful.

A cautious final formulation is:

> The data do not prove the whole corpus. They support one empirical implication compatible with it: global growth remains materially and energetically conditioned, even when national accounts make that materiality less visible.

Online sources and references

[1] Pierre Campergue, *Reflections on the Economic Power of Work and the Mechanical Power of Money*, ResearchGate page.
https://www.researchgate.net/publication/374034279_Reflections_on_the_economic_power_of_Work_and

[2] Pierre Campergue and Mélik Lemariey, *Liminal Insights: Bridging Economics and Thermodynamics*, ResearchGate page.
https://www.researchgate.net/publication/373783021_Liminal_Insights_Bridging_Economics_and_Thermo

[3] World Bank, World Development Indicators metadata, `NY.GDP.MKTP.KD`, GDP constant market U.S. dollars.
https://databank.worldbank.org/metadataglossary/world-development-indicators/series/NY.GDP.MKTP.KD

[4] World Bank, World Development Indicators metadata, `NY.GDP.MKTP.PP.KD`, GDP PPP constant international dollars.
https://databank.worldbank.org/metadataglossary/world-development-indicators/series/NY.GDP.MKTP.PP.KD

[5] World Bank, World Development Indicators metadata, `EG.GDP.PUSE.KO.PP.KD`, GDP per unit of energy use.
https://databank.worldbank.org/metadataglossary/world-development-indicators/series/EG.GDP.PUSE.KO.PP.KD

[6] World Bank Data Help Desk, API basic call structures.
https://datahelpdesk.worldbank.org/knowledgebase/articles/898581-api-basic-call-structures

[7] International Energy Agency, Energy Statistics Data Browser.
https://www.iea.org/data-and-statistics/data-tools/energy-statistics-data-browser

[8] International Energy Agency, World Energy Balances.
https://www.iea.org/data-and-statistics/data-product/world-energy-balances

[9] David I. Stern, "Energy and economic growth in the USA: A multivariate approach", *Energy Economics*, 1993.
https://www.sciencedirect.com/science/article/pii/014098839390033N

[10] James E. Payne, "A survey of the electricity consumption-growth literature", *Applied Energy*, 2010.
https://ideas.repec.org/a/eee/appene/v87y2010i3p723-731.html

[11] Ilhan Ozturk, "A literature survey on energy-growth nexus", *Energy Policy*, 2010.
https://ideas.repec.org/a/eee/enepol/v38y2010i1p340-349.html

[12] Zsuzsanna Csereklyei, Maria del Mar Rubio-Varas, and David I. Stern, "Energy and economic growth: the stylized facts".
https://researchportalplus.anu.edu.au/en/publications/energy-and-economic-growth-the-stylized-facts/

[13] Helmut Haberl et al., "A systematic review of the evidence on decoupling of GDP, resource use and GHG emissions".
https://research.ceu.edu/en/publications/a-systematic-review-of-the-evidence-on-decoupling-of-gdp-resource-2/

[14] University of Montpellier, "At the global scale, no decoupling to expect between GDP and energy consumption".
https://www.umontpellier.fr/en/articles/a-lechelle-mondiale-aucun-decouplage-a-attendre-entre-pib-et-consommation-denergie

[15] UK Energy Research Centre, "Energy efficiency contributed 25% of UK economic growth since 1971".
https://ukerc.ac.uk/news/energy-efficiency-contributed-25-of-uk-economic-growth-since-1971/

[16] Bank of England, "Money in the modern economy: an introduction", Quarterly Bulletin 2014 Q1.
https://www.bankofengland.co.uk/quarterly-bulletin/2014/q1/money-in-the-modern-economy-an-introduction

[17] Bank of England, "Money creation in the modern economy", Quarterly Bulletin 2014 Q1.
https://www.bankofengland.co.uk/quarterly-bulletin/2014/q1/money-creation-in-the-modern-economy

[18] Bank of England, "What is money?"
https://www.bankofengland.co.uk/explainers/what-is-money

[19] European Central Bank, "What is money?"
https://www.ecb.europa.eu/ecb-and-you/explainers/tell-me-more/html/what_is_money.en.html

[20] Augusto Graziani, *The Monetary Theory of Production.*
https://philpapers.org/rec/GRATMT-7

[21] Geoffrey Ingham, “The Nature of Money”, working paper / lecture text.
https://www.economix.fr/uploads/source/doc/colloques/2009_cartelier/Ingham.pdf

---

Disclaimer

This working paper is part of a repository research process. It is not a peer-reviewed article. It is intended to make claims inspectable, reproducible, and open to criticism. Errors, interpretations, and empirical extensions are the responsibility of the author of this article, not Pierre Campergue.